



JOB SHEET 6

SELECTION STATEMENTS 2

1. Objectives

1. Students can solve problems and case studies using nested selection statements.
2. Students can apply nested selection statements in Java programs.
3. Students can apply the logical operators &&, ||, and ! in selection structures.

2. Labs Activities

2.1 Experiment 1: Nested IF to Check Thesis Exam Requirements

Time: 90 minutes

A student wants to register for the thesis exam. The SIMTA system first checks an administrative requirement: the student must have no outstanding penalties. If this requirement is met, the system then checks the guidance log. To register for the exam, the student must have at least 8 guidance sessions with Supervisor 1 and at least 4 guidance sessions with Supervisor 2. If all requirements are met, the student can proceed to register for the thesis exam. If not, the system shows the reason for failure. Based on this case, build the Java program using the steps below.

1. Create a new file named **NestedThesisExamAttendanceNo.java** inside the **jobsheet6** folder
2. Build the basic Java program structure with a **main()** function
3. Import the **Scanner** library and declare a scanner named **sc**
4. Declare a **String** variable named **message** to store the output message

```
String message;
```

5. Add the following code to read keyboard input about the penalty status, storing the result in a **String** variable named **noPenalty**

```
System.out.print("Has the student cleared all penalties? (Yes/No): ");  
String noPenalty = sc.nextLine().trim();
```

Note: the **trim()** function removes leading and trailing spaces from the **String**

6. Add the following code to read keyboard input about the guidance log, storing the results in **int** variables named **guidanceCount1** and **guidanceCount2**

```
System.out.print("Enter the number of guidance sessions with Supervisor 1: ");  
int guidanceCount1 = sc.nextInt();
```



```
System.out.print("Enter the number of guidance sessions with Supervisor 2: ");
int guidanceCount2 = sc.nextInt();
```

7. Build a Nested-IF selection structure that checks the penalty condition at the first level and the guidance session counts at the second level

```
if (noPenalty.equalsIgnoreCase("Yes")) {
    if (guidanceCount1 >= 8 && guidanceCount2 >= 4) {
        message = "All requirements met. The student may register for the thesis exam";
    } else if (guidanceCount1 < 8 && guidanceCount2 < 4) {
        message = "Failed! Guidance sessions with Supervisor 1 are below 8 and Supervisor 2 are below 4";
    } else if (guidanceCount1 < 8) {
        message = "Failed! Guidance sessions with Supervisor 1 have not reached 8";
    } else {
        message = "Failed! Guidance sessions with Supervisor 2 have not reached 4";
    }
} else {
    message = "Failed! The student still has an outstanding penalty";
}
System.out.println(message);
```

8. Compile and run the program so that it produces the following output.

```
Has the student cleared all penalties? (Yes/No): yes
Enter the number of guidance sessions with Supervisor 1: 6
Enter the number of guidance sessions with Supervisor 2: 5
Failed! Guidance sessions with Supervisor 1 have not reached 8
```

Questions

1. What happens if the student answers "No" to the penalty-clearance question? Why?
2. Explain the meaning of the following code snippet!

```
if (guidanceCount1 >= 8 && guidanceCount2 >= 4) {
```

3. Describe the full flow of checking the student's requirements from start to finish. Explain step by step for every condition!

2.2 Experiment 2: Logical Operators to Determine Campus WiFi Access

Time: 50 minutes

The campus WiFi can only be used by students or lecturers whose accounts are not blocked. The program receives information on whether the user is a student, whether the user is a lecturer, and whether the user's account is currently blocked. Access is granted if the user is a student or a lecturer, and the account is not blocked. This experiment practices the logical operators **&&** (AND), **||** (OR), and **!** (NOT).

1. Create a new file named **LogicalOperatorWifiAttendanceNo.java** inside the **jobsheet6** folder.
2. Build the basic Java program structure with a **main()** function.
3. Import the **Scanner** library and declare a scanner named **sc**.
4. Declare three **boolean** variables: **isStudent**, **isLecturer**, and **isBlocked**.

```
boolean isStudent;
boolean isLecturer;
boolean isBlocked;
```

5. Add the following code to read input from the keyboard.

```
System.out.print("Is the user a student? (true/false): ");
isStudent = sc.nextBoolean();

System.out.print("Is the user a lecturer? (true/false): ");
isLecturer = sc.nextBoolean();

System.out.print("Is the account currently blocked? (true/false): ");
isBlocked = sc.nextBoolean();
```

6. Build the following selection structure to determine WiFi access.

```
if ((isStudent || isLecturer) && !isBlocked) {
    System.out.println("WiFi access granted");
} else {
    System.out.println("WiFi access denied");
}
```

7. Compile and run the program. Test it with the following input combinations and observe the results.

Test	isStudent	isLecturer	isBlocked
1	true	false	false
2	false	true	false
3	true	false	true
4	false	false	false

Questions

1. Explain the function of the **||**, **&&**, and **!** operators in the condition above.
2. Why can a lecturer still get access when **isStudent = false**?
3. Change **||** to **&&**. Run the program again using test data 1 and 2. What happens, and why?
4. In the expression **isStudent || isLecturer**, when does **isLecturer** not need to be evaluated? Explain using short-circuit evaluation.
5. In the expression **(isStudent || isLecturer) && !isBlocked**, when does **!isBlocked** not need to be evaluated? Explain.



2.3 Experiment 3: Nested IF and Logical Operators to Determine Laboratory Access

Time: 60 minutes

A student may use the laboratory outside class hours if their status is active and they are not currently under sanction. If this requirement is met, the system performs a second check. Laboratory access is granted if the student has lecturer permission or is a lab assistant. This case combines nested selection with logical operators.

1. Create a new file named **NestedLabAccessAttendanceNo.java** inside the **jobsheet6** folder.
2. Build the basic Java program structure with a **main()** function.
3. Import the **Scanner** library and declare a scanner named **sc**.
4. Declare the **boolean** variables **isActiveStudent**, **isSanctioned**, **hasLecturerPermit**, and **isLabAssistant**.

```
boolean isActiveStudent;  
boolean isSanctioned;  
boolean hasLecturerPermit;  
boolean isLabAssistant;
```

5. Add code to read all four variables from the keyboard using **sc.nextBoolean()**.
6. Build the Nested IF structure with logical operators as shown below.

```
if (isActiveStudent && !isSanctioned) {  
    if (hasLecturerPermit || isLabAssistant) {  
        System.out.println("Laboratory access granted");  
    } else {  
        System.out.println("Access denied: lecturer permission or lab assistant  
status required");  
    }  
} else {  
    System.out.println("Access denied: student status does not meet the  
requirement");  
}
```

7. Compile and run the program. Test at least four input combinations so that every possible output occurs at least once.
8. Record the input combinations you used and their outputs.

Questions

1. Why is the check **hasLecturerPermit || isLabAssistant** placed inside the first IF?
2. Explain the function of the **&&**, **||**, and **!** operators in this program.



3. Can the access requirement be written as a single condition: **`isActiveStudent && !isSanctioned && (hasLecturerPermit || isLabAssistant)`**? Explain whether the final access decision stays the same.
4. What is the advantage of using Nested IF in this case, compared to a single IF, if the system needs to show different reasons for denial?
5. Create one input combination that causes access to be denied at the first level, and one that causes it to be denied at the second level.

3. Assignment

Time: 120 minutes

1. Implement the flowchart you created in Exercise 2 of Week 6 for the bookstore discount system as a Java program. The program must use nested selection statements (Nested IF). Use logical operators where needed.
2. Write a Java program for a lab-assistant candidate selection system based on the following rules:
 - a. A student may take part in the selection if their status is active and they are not currently under academic sanction.
 - b. If this requirement is met, the student must also meet the next requirement: a minimum grade of 80 in Basic Programming, or a programming competency certificate.
 - c. If both requirements are met, the student will be called for an interview. The student is accepted as an assistant if the interview score is at least 75.
 - d. The program must show the reason if the student fails at any stage of the selection.
 - e. Use nested selection and logical operators. Save the file as **`Task2AssistantSelectionAttendanceNo.java`**.